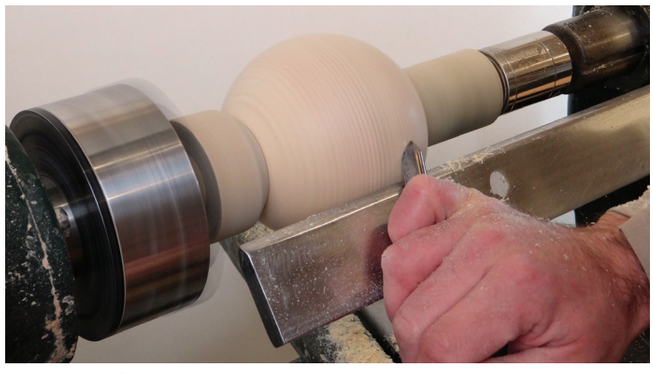
The sphere held in the wooden cups by its equator, the poles are straight up and down

The grain of the sphere in this picture shows clearly which way it wants to be cut, but if you mount it diagonally you will always be working against the grain at two points in each rotation

With the cups turned the sphere can be remounted. It is important to understand the grain direction of the sphere before starting to turn it. You need to ensure the poles of the globe are directly up and down, or north and south, to ensure a clean cut. When turning the blank initially, it was a standard spindle blank, so the cuts ran from the equator, down to the poles. Now, with the blank mounted between cups, it becomes faceplate work, so the cuts will be drawn apparently uphill, but still from the equator to the poles, and so still working with the grain of the wood. What you can't do at this stage, is to mount the sphere with the poles running diagonally because then you won't be cutting consistently with the grain; at some point you will be working against the grain. I'm not one to say this will never work, but it will be making life difficult for yourself when it comes to achieving a good, even surface finish.

Turning between cup centres



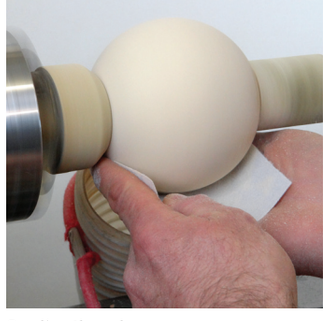
Use the wing of the tool in a shear cutting action to achieve the best results

You can see the tooling marks on the sphere, which shows how this method automatically corrects any imperfections in your freehand sphere turning

With the sphere held at the equator by the cups, and the poles running directly vertically, you can finish turn the sphere. Light cuts with the wing of the tool in a shearing cut work really well here. This is when it becomes clear just how simple this all is, because as you turn in this new orientation, the waste marks will be removed and, because the lathe simply spins the sphere, you will automatically be removing anything that wasn't perfectly spherical from the earlier operation. Because you are using a light shearing cut, working with the grain of the wood, it cuts cleanly and a sphere is formed almost effortlessly. Check the sphere regularly and reposition it in the cups occasionally, but only by rotating it around the equator, so a different portion of the sphere can be turned. Once it is all turned and the cut off marks are gone, the sphere is ready for sanding.

The completed spheres

The sanding process is even more simple. I worked from 120 grit to 240 grit on these softwood spheres, but you might choose to go finer. You can sand in any direction, so now you can feel free to mount the ball in whatever orientation you like between the cups; just try to ensure every surface is worked evenly. Very quickly, you will find you have a perfectly smooth and spherical ball. ●



Sanding the sphere

The completed spheres

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